

Undistortion

Limits

⌚ Time

2000 ms

📁 Memory

512 MiB

<> Source size

64 KiB

☑ Points

100

Submit

Code

C++

▼

Submit

Verdict	Time	Memory	Language	Points	Samples
No submissions yet :(					

Description

You are given a point $P_u(x_u, y_u)$ in 2D coordinate system and two positive real numbers k_1 and k_2 . The numbers k_1 and k_2 are called the **distortion parameters**. With those values, we construct a point $P_d(x_d, y_d)$ with the following formula:

$$r^2 = x_u^2 + y_u^2 \quad s = 1.0 + k_1 r^2 + k_2 r^4 \quad x_d = s \cdot x_u, \quad y_d = s \cdot y_u$$

This is called the distortion formula and point P_d is called the **distorted point** of P_u . Similarly, P_u is called the **undistorted point** of P_d .

Below is a short code sample how the formula would look in C++.

```
pair<float, float> Distort(float k1, float k2, float xu, float yu){
    float r2 = xu * xu + yu * yu;
    float s = 1.0 + r2 * k1 + r2 * r2 * k2;
```

```
    return make_pair(xu * s, yu * s);  
}
```

Given several examples of k_1 , k_2 and $P_d(x_d, y_d)$, for each of them find the corresponding $P_u(x_u, y_u)$.

In other words, given the distortion parameters and the distorted point, determine the corresponding undistorted point P_u which maps to the given P_d by the above formula.

Round the coordinates of P_u to three decimal places. If your output differs from the official output by at most 0.001, your output will be considered correct.

Input Data

In the first line, there is an integer $3 \leq T \leq 40000$, the number of test cases.

In each of the next T lines there are four real numbers k_1, k_2, x_d, y_d in that order, with up to three decimal places. They represent the distortion parameters and the coordinates of the distorted point. It will be $0 \leq k_1, k_2 \leq 2.0$ and $0 \leq x_d, y_d \leq 15.0$.

Output Data

Print T lines, one for each input test case. In each line, print two real numbers x_u and y_u , the coordinates of the undistorted point corresponding to that test case, rounded to three decimal places.

Scoring

First test data cluster is worth 10 points and $T \leq 10$. Second test data cluster is worth 20 points and $T \leq 5000$. Third test data cluster is worth 70 points and there are no additional constraints.

Example

Input:

```
2  
0.05 1.87 1.016 1.181  
0.84 1.48 0.417 0.655
```

Output:

```
0.537 0.624  
0.298 0.468
```

Explanation:

For the first line, the point $(0.537, 0.624)$ maps to $(1.016, 1.181)$ with $k_1 = 0.05$ and $k_2 = 1.87$, so that is the answer (everything up to 3 decimals).

For the second line, the point $(0.298, 0.468)$ maps to $(0.417, 0.655)$ with $k_1 = 0.84$ and $k_2 = 1.48$, so that is the answer (everything up to 3 decimals).